

LESSON 8

THE CONDITIONAL AND RELATED STATEMENTS

Overview:

This lesson will explore the equivalent forms of conditional statements. The lesson will start with introducing how a conditional statement can be written in an equivalent form. The related statements such as the converse, inverse, and contrapositive will be included in this portion. This will then be followed by determining the truth value of these statements.

Learning Outcomes:

At the end of lesson 8, the students can:

1. Identify the different forms of conditional statement;
2. write a conditional statement to its equivalent form; and
3. state the inverse, converse, and contrapositive of a given conditional statement.

Materials Needed:

- learning module
- video lesson

Duration: 1.5 hours

Learning Contents:

8.1 THE CONDITIONAL STATEMENT

The conditional statement $p \rightarrow q$ is not only written in the form “If p , then q .”. For example, the statement “If it is a holiday (p), then I have no class (q).”, can also be stated as “I have no class (q), if it is holiday (p).”. Observe that the antecedent (p) comes first before its consequent (q) in the statement “If it is a holiday (p), then I have no class (q).”. However, in the statement “I have no class (q), if it is holiday (p).”, the consequent (q) has been mentioned first. This is just one of the different forms of conditional statement.

The following table shows the list of other forms of conditional statements.

Table 8.1 Common Forms of $p \rightarrow q$

Every conditional statement $p \rightarrow q$ can be written in the following forms.	
If p , then q .	Every p is a q .
If p , q .	q , if p .
p only if q .	q provided that p .
p implies q .	p is a necessary condition for q .
Not p or q .	p is a sufficient condition for q .

Examples

Write each of the following in “If p , then q .” form.

- 1. Tomorrow is Tuesday, only if today is Monday.
- 2. The number is divisible by 10 provided that it ends with a 0.

Solution

- 1. The statement “Tomorrow is Tuesday, only if today is Monday.”, is in “ p only if q .” form. The antecedent is “Tomorrow is Tuesday.” and the consequent is “Today is Monday.”. Thus its “If p , then q .” form is “If tomorrow is Tuesday, then today is Monday.”.
- 2. The statement “The number is divisible by 10 provided that it ends with a 0.”, is in “ q provided that p .” form. The antecedent is “It ends with a 0.” and the consequent is “The number is divisible by 10.”. Thus its “If p , then q .” form is “If the number ends with a 0, then it is divisible by 10”.

An Equivalent Form of the Conditional (Switcheroo)

$$p \rightarrow q \equiv \sim p \vee q$$

Example

- 3. Restate the conditional statement “If you remember your Facebook account’s password, then you will be able to log in.” to it equivalent disjunctive form.

Solution

In this case, we will write the disjunction of the negation of the antecedent and copy the consequent. Hence, its equivalent disjunctive form is “You did not remember your Facebook account’s password or you will be able to log in.”.

The Negation of the Conditional

$$\sim (p \rightarrow q) \equiv p \wedge \sim q$$

Example

4. Given the previous example of conditional statement “If you remember your Facebook account’s password, then you will be able to log in.”, write its negation.

Solution

We write the conjunction of the antecedent and the negation of the consequent. Hence, its negation form is “You remember your Facebook account’s password or you will not be able to log in.”.

8.2 THE CONVERSE, THE INVERSE, AND THE CONTRAPOSITIVE

Each conditional statement has three related statements. These statements are called the *converse*, the *inverse*, and the *contrapositive*.

The **converse** of $p \rightarrow q$ is $q \rightarrow p$.

The **inverse** of $p \rightarrow q$ is $\sim p \rightarrow \sim q$.

The **contrapositive** of $p \rightarrow q$ is $\sim q \rightarrow \sim p$.

Example

5. Write the converse, inverse, and contrapositive of “If I study hard, then I will pass the test.”.

Solution

Converse: If I pass the test, then I study hard.

Inverse: If I do not study hard, then I will not pass the test.

Contrapositive: If I do not pass the test, then I did not study hard.

The following table shows the truth tables for conditional and its related statements. Based on the table, we can say that every conditional statement is equivalent to its contrapositive and that the converse and inverse are also equivalent with each other.

Table 8.2 Truth Tables for Conditional and Related Statements

P	q	Conditional $p \rightarrow q$	Converse $q \rightarrow p$	Inverse $\sim p \rightarrow \sim q$	Contrapositive $\sim q \rightarrow \sim p$
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

$q \rightarrow p \equiv \sim p \rightarrow \sim q$

$p \rightarrow q \equiv \sim q \rightarrow \sim p$

We can use the table to determine whether two statements are equivalent or not.

Example

Determine whether the given statements are equivalent.

6. If the weather is cold, then we wear a sweater.

If we wear a sweater, then the weather is cold.

7. If a person is yawning, then he is sleepy.

If a person is not sleepy, then he is not yawning.

Solution

6. The second statement is the converse of the first. Hence, the statements are not equivalent.
7. The second statement is the contrapositive of the first. Thus, the statements are equivalent.

Also, we can determine the truth value of a conditional statement using its contrapositive or vice versa.

Example

Write the contrapositive of each statement and use the contrapositive to determine whether the original statement is true or false.

8. If a number is divisible by 2, then the number is an even number.

9. If two lines intersect, then the lines are not parallel.

Solution

8. If a number is not an even number, then the number is not divisible by 2.
This is a true statement. Hence, the original statement is also true.
9. If two lines are parallel, then the lines will not intersect. This statement is true and so is the original statement.

Learning Activity:

Divide the class into four groups. Each of the four groups will have its representative to answer every question on the board simultaneously. There will be a total of ten (10) questions with different time allotments depending on the difficulty of the problem.

Questions:

1. Write the statement “Every odd prime number is greater than 2.” in “If p , then q .” form.

2. Write the statement "*He can join the band, if he has the talent to play a keyboard.*" in "If p , then q ." form.
3. Write the statement "*Being in excellent shape is a necessary condition for running a marathon.*" in "If p , then q ." form.
4. Give the converse of the statement "*If I were rich, I would quit this job.*".
5. What is the inverse of the statement "*If we had a car, we would be able to attend the meeting on time.*".
6. Write the contrapositive of the statement "*If you are 18, you can live independently.*".
7. What is the converse of the inverse of the contrapositive of $p \rightarrow q$?
8. Express the conditional statement "*If $3x + 7 = 11$, then $x \neq 7$.*" to its equivalent disjunctive form.
9. Give the negation of the conditional statement "*If $x \neq 3$, then $5x + 7 \neq 22$.*".
10. Give the contrapositive of the statement "*If $a + b$ is divisible by 3, then a is divisible by 3 and b is divisible by 3.*" Use the contrapositive to determine whether the original statement is true or false.

Learning Evaluation:

- A. Write each statement in "If p , then q ." form.
 1. Farah got the highest score, if she studied alone.
 2. You can enroll next semester provided that you passed all of your subjects.
 3. You are not staying home or you have to do household chores.
 4. Every number that ends with a 3 is divisible by 3.
- B. Write the converse, inverse, and contrapositive of the "If p , then q ." forms from part A item numbers 1 to 3.
- C. Using "If p , then q ." form of part A item number 4, write its contrapositive. Determine whether the original statement is true or false using its contrapositive.

References:

- Aufman, R. et. al. (2018). *Mathematics in the Modern World*. Rex Book Store Inc. Sampiloc, Manila, Philippines.
- Johnstone, P. (1987). *Mathematics, Logic, Categories, and Sets*. Cambridge University Press.